SYSDESIGN REFERENCE

Database Replication Tradeoff Matrix

A decision-oriented comparison of alternatives and failure modes.

Copy data across nodes to improve read scale and resilience while managing lag and failover.

VERSION 1.0 REVIEW

REVIEWED THROUGH 2026-07-18

STATUS HUMAN REVIEW REQUIRED

SYSDESIGN / DATABASE-REPLICATION

SYSDESIGN

Database Replication Tradeoff Matrix

CORE IDEA

Copy data across nodes to improve read scale and resilience while managing lag and failover.

Reference architecture decision chain

01 Leader/Follower Vs Multi-Leader	02 Sync Level	03 Read Routing	04 And Recovery Point.

Decisions to defend

+ Leader/follower vs multi-leader
+ sync level
+ read routing
+ and recovery point.

Failure modes to test

+ Replication lag
+ split brain
+ lost writes
+ and failover storms.

Interview flow

- [ ] Clarify users, critical paths, and non-goals
- [ ] Estimate only scale that changes a decision
- [ ] Define API, data model, and source of truth
- [ ] Draw synchronous and asynchronous paths
- [ ] Pressure-test timeout, retry, overload, and recovery
- [ ] Close with tradeoffs and one alternative

Deep-dive drill

Database Replication Tradeoff Matrix

Probe	What a strong answer makes explicit
How does the design behave during a regional dependency failure?	Requirement, decision, cost, failure behavior, and observable metric.
Which metric reveals the first bottleneck?	Requirement, decision, cost, failure behavior, and observable metric.
What consistency does the user actually observe?	Requirement, decision, cost, failure behavior, and observable metric.
How would the design change at ten times the traffic?	Requirement, decision, cost, failure behavior, and observable metric.

Data questions

+ Partition key
+ Versioning
+ Uniqueness
+ Retention
+ Reconciliation

Operations questions

+ SLO
+ Backlog
+ Capacity
+ Alert
+ Failover test

DESIGN DISCIPLINE

For every box: which requirement justifies it, what cost does it introduce, and what happens when it fails?

PUBLISHING STATUS
REVIEW GATE: TECHNICAL REVIEW. AI-produced draft. A named human must verify technical accuracy, sourcing, freshness, and editorial quality before publication. Reviewed-through date: 2026-07-18.